The factorized Pauli–Villars multigrid of RutticeQCD, from the solver to HMC: a step-by-step walkthrough

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October 23, 2026 (internal note)

This document derives, step by step, what the MG note[1] ("the note") states. The note gives the method and its results in compact form; here every formula is taken apart line by line, with every symbol defined before it is used, so that the method can be followed with nothing more than standard graduate coursework in lattice QCD. The sections follow the note: each heading names the section of the note it derives (Secs. IV–XI), so the two can be read side by side. The last two sections go beyond the note: the solves of HMC, derived from the path integral (the note only reports the HMC measurements, Sec. XII.G), and a map from each step to the code. Nothing here is new with respect to the note and the code.

Contents

1 The operator and Pauli–Villars preconditioning (note Sec. IV) 2
- 1.1 Ingredients . . . 2
- 1.2 The free spectrum of W . . . 3
- 1.3 The operator in Brower–Neff–Orginos form (note Sec. IV.A) . . . 3
- 1.4 Splitting off the fifth-dimension hop . . . 3
- 1.5 The Γ_5-transformation (note Sec. IV.B) . . . 4
- 1.6 The site-diagonal part C and its inverse (note Sec. IV.C) . . . 5
- 1.7 The Pauli–Villars preconditioned operator (note Sec. IV.D) . . . 6
- 1.8 Neff's α (note Sec. IV.F) . . . 7

2 Even-odd forms (note Sec. V) 7
- 2.1 Blocks . . . 7
- 2.2 The Schur complement . . . 7
- 2.3 $S^{-1} = \pi D^{-1} \iota$ . . . 7
- 2.4 The Pauli–Villars preconditioned forms . . . 8

3 The factorized coarse-grid correction and the prolongation (note Sec. VI) 8
- 3.1 The factorized coarse correction . . . 8
- 3.2 Aggregation and the chirality split . . . 10
- 3.3 Orthonormalization per block: CholeskyQR2 . . . 10
- 3.4 The basis $\{s, \hat{t}\}$ and why the correction is exact . . . 11
- 3.5 The coarse operators by probing . . . 12

[1] `docs/mg_background/MDWF_MULTIGRID_NOTE.tex` in the RutticeQCD repository.

4 The coarse solvers (note Sec. VII) 12

5 Setup (note Sec. VIII) 13
5.1 Chebyshev-filtered subspace iteration . . . 14
5.2 Bootstrap . . . 14
5.3 Cost and memory . . . 14

6 The Chebyshev smoother and the cycle (note Sec. IX) 15
6.1 The Chebyshev smoother, derived . . . 15

7 The outer solver: flexible GCR (note Sec. X) 16

8 Solving with $D^\dagger$ (note Sec. XI) 16
8.1 $\Gamma_5 D^{-1}\Gamma_5$ as a preconditioner of $D^\dagger$ (note Sec. XI.A) . . . 16
8.2 On the even sites, step by step (note Secs. XI.B–C) . . . 17
8.3 Why D_- cannot be inverted (note Sec. XI.D) . . . 17

9 HMC 17
9.1 What we want to compute . . . 18
9.2 Pseudofermions: a determinant as a Gaussian integral . . . 18
9.3 Even-odd: only K depends on the links . . . 19
9.4 The idea of HMC . . . 20
9.5 The integrator . . . 20
9.6 The heat bath for ϕ_e . . . 21
9.7 The force . . . 22
9.8 How precisely to solve . . . 22
9.9 Solving $(K^\dagger K)^{-1}$ . . . 22
9.10 What the multigrid keeps along a trajectory . . . 23
9.11 Memory (24^4, $m = 0.001$, 20 vectors, levels below, A30) . . . 23
9.12 What the measurements say so far . . . 24

10 Map to the code 24

1 The operator and Pauli–Villars preconditioning (note Sec. IV)

1.1 Ingredients

The field ψ lives on the four-dimensional lattice and carries a fifth index $s = 0, \dots, L_s - 1$; each ψ_s is a Dirac field. The chiral projectors are

$$P_\pm = \frac{1 \pm \gamma_5}{2}, \qquad P_\pm^2 = P_\pm, \qquad P_+ P_- = 0, \qquad P_+ + P_- = 1. \tag{1.1}$$

The Wilson hopping term and the Wilson operator of mass $-M_5$ are

$$(K\psi)(x) = \sum_{\mu=1}^{4} \Big[(1 - \gamma_\mu) U_\mu(x) \psi(x + \hat{\mu}) + (1 + \gamma_\mu) U_\mu^\dagger(x - \hat{\mu}) \psi(x - \hat{\mu}) \Big], \tag{1.2}$$

$$W = (4 - M_5) - \tfrac{1}{2} K. \tag{1.3}$$

W acts on every slice s alike and does not mix slices.

1.2 The free spectrum of W

The spectrum of W on the free field ($U_\mu = 1$) is needed twice: for the spectra of the note (Sec. III) and for D_- (Sec. 8.3). Take a plane wave $\psi(x) = e^{ip\cdot x}u$ with a constant spinor u:

$$(K\psi)(x) = \sum_\mu \Big[(1-\gamma_\mu)e^{ip_\mu} + (1+\gamma_\mu)e^{-ip_\mu}\Big]\psi(x) \tag{1.4}$$

$$= \sum_\mu \Big[(e^{ip_\mu} + e^{-ip_\mu}) - \gamma_\mu(e^{ip_\mu} - e^{-ip_\mu})\Big]\psi(x) \tag{1.5}$$

$$= \sum_\mu \Big[2\cos p_\mu - 2i\gamma_\mu \sin p_\mu\Big]\psi(x), \tag{1.6}$$

$$W\psi = \Big[(4 - M_5) - \sum_\mu \cos p_\mu + i\sum_\mu \gamma_\mu \sin p_\mu\Big]\psi \tag{1.7}$$

$$= \Big[- M_5 + \sum_\mu (1-\cos p_\mu) + i\sum_\mu \gamma_\mu \sin p_\mu\Big]\psi. \tag{1.8}$$

The matrix $G = \sum_\mu \gamma_\mu \sin p_\mu$ is Hermitian, and since the γ_μ anticommute,

$$G^2 = \sum_{\mu,\nu} \gamma_\mu\gamma_\nu \sin p_\mu \sin p_\nu = \sum_\mu \gamma_\mu^2 \sin^2 p_\mu + \sum_{\mu<\nu}\{\gamma_\mu,\gamma_\nu\}\sin p_\mu \sin p_\nu = \sum_\mu \sin^2 p_\mu, \tag{1.9}$$

so its eigenvalues are $\pm(\sum_\mu \sin^2 p_\mu)^{1/2}$. The eigenvalues of W are therefore

$$w(p) = -M_5 + \sum_\mu (1-\cos p_\mu) \pm i\Big(\sum_\mu \sin^2 p_\mu\Big)^{1/2}. \tag{1.10}$$

Their real parts run from $-M_5$ ($p = 0$) to $8 - M_5$ ($p_\mu = \pi$); the zeros of the real part at small $|p|$ are the physical domain-wall modes.

1.3 The operator in Brower–Neff–Orginos form (note Sec. IV.A)

With b, c the Möbius parameters, define

$$D_+ = 1 + bW, \qquad D_- = cW - 1. \tag{1.11}$$

The operator at quark mass m acts slice by slice as

$$(D\psi)_s = D_+\psi_s + D_-(P_-\psi_{s+1} + P_+\psi_{s-1}), \qquad \psi_{L_s} \equiv -m\psi_0, \qquad \psi_{-1} \equiv -m\psi_{L_s-1}. \tag{1.12}$$

For $L_s = 4$ the matrix in s is

$$D = \begin{pmatrix} D_+ & D_-P_- & 0 & -mD_-P_+ \\ D_-P_+ & D_+ & D_-P_- & 0 \\ 0 & D_-P_+ & D_+ & D_-P_- \\ -mD_-P_- & 0 & D_-P_+ & D_+ \end{pmatrix}. \tag{1.13}$$

The physical quark field is $q = P_-\psi_0 + P_+\psi_{L_s-1}$; the runs use $b = 1.5$, $c = 0.5$, $L_s = 8$.

1.4 Splitting off the fifth-dimension hop

Define the hop in the fifth dimension, with a minus sign,

$$(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1} \qquad \text{(same wrap-around)}. \tag{1.14}$$

Comparing with Eq. (1.12), the off-diagonal part of D is $-D_-Y$, with D_- on the *left*:

$$D = D_+ - D_-Y. \tag{1.15}$$

Now insert $D_\pm$ and then W, one substitution per line:

$$D = (1 + bW) - (cW - 1)Y \tag{1.16}$$
$$= 1 + bW - cWY + Y \tag{1.17}$$
$$= 1 + Y + W\,(b - cY) \tag{1.18}$$
$$= 1 + Y + \Big[(4 - M_5) - \tfrac{1}{2}K\Big](b - cY) \tag{1.19}$$
$$= \underbrace{\big[1 + b(4 - M_5)\big]}_{d_+} + \underbrace{\big[1 - c(4 - M_5)\big]}_{d_-} Y - \tfrac{1}{2}K\,(b - cY) \tag{1.20}$$
$$= C - \tfrac{1}{2}K\,(b - cY), \qquad C \equiv d_+ + d_- Y. \tag{1.21}$$

C contains no four-dimensional hop: it acts within one site, as an $L_s \times L_s$ matrix in s. All the gauge field sits in $-\frac{1}{2}K(b - cY)$. The Pauli–Villars operator is the same operator at unit mass,

$$D_{\rm PV} = D\big|_{m=1}, \qquad C_{\rm PV} = C\big|_{m=1}. \tag{1.22}$$

1.5 The Γ_5-transformation (note Sec. IV.B)

For the Wilson operator $\gamma_5 W \gamma_5 = W^\dagger$. Domain-wall fermions use $\Gamma_5 = \gamma_5 R_5$, $(R_5\psi)_s = \psi_{L_s-1-s}$. We derive how each block transforms.

K. Read off the matrix elements of K between neighbours:

$$K_{x,x+\hat\mu} = (1 - \gamma_\mu)U_\mu(x), \qquad K_{x+\hat\mu,x} = (1 + \gamma_\mu)U_\mu^\dagger(x). \tag{1.23}$$

The adjoint swaps the indices and conjugates each block (spin and colour commute, and the γ_μ are Hermitian):

$$(K^\dagger)_{x,x+\hat\mu} = \big(K_{x+\hat\mu,x}\big)^\dagger = U_\mu(x)\,(1 + \gamma_\mu) = (1 + \gamma_\mu)U_\mu(x), \tag{1.24}$$
$$(K^\dagger)_{x+\hat\mu,x} = \big(K_{x,x+\hat\mu}\big)^\dagger = (1 - \gamma_\mu)U_\mu^\dagger(x). \tag{1.25}$$

So $K^\dagger$ is K with $\gamma_\mu \to -\gamma_\mu$. Since $\gamma_5\gamma_\mu\gamma_5 = -\gamma_\mu$,

$$\gamma_5 K \gamma_5 = K^\dagger, \qquad \gamma_5 W \gamma_5 = W^\dagger. \tag{1.26}$$

W acts on every slice alike, so it commutes with R_5: $\Gamma_5 W \Gamma_5 = W^\dagger$, and the same for $D_\pm =$ functions of W: $\Gamma_5 D_\pm \Gamma_5 = D_\pm^\dagger$.

Y. Write Y as a matrix in s ($L_s = 4$), from $(Y\psi)_s = -P_-\psi_{s+1} - P_+\psi_{s-1}$ and the wrap-around $\psi_{L_s} = -m\psi_0$, $\psi_{-1} = -m\psi_{L_s-1}$:

$$Y = \begin{pmatrix} 0 & -P_- & 0 & mP_+ \\ -P_+ & 0 & -P_- & 0 \\ 0 & -P_+ & 0 & -P_- \\ mP_- & 0 & -P_+ & 0 \end{pmatrix}. \tag{1.27}$$

γ_5 commutes with $P_\pm$, so $\Gamma_5 Y \Gamma_5 = R_5 Y R_5$, whose entry (s, s') is $Y_{L_s-1-s,\,L_s-1-s'}$: the matrix rotated by 180°. Compare with $Y^\dagger$, whose entry (s, s') is $(Y_{s',s})^\dagger$:

$$(R_5 Y R_5)_{0,1} = Y_{3,2} = -P_+, \qquad (Y^\dagger)_{0,1} = (Y_{1,0})^\dagger = -P_+, \tag{1.28}$$
$$(R_5 Y R_5)_{1,0} = Y_{2,3} = -P_-, \qquad (Y^\dagger)_{1,0} = (Y_{0,1})^\dagger = -P_-, \tag{1.29}$$
$$(R_5 Y R_5)_{0,3} = Y_{3,0} = mP_-, \qquad (Y^\dagger)_{0,3} = (Y_{3,0})^\dagger = mP_-, \tag{1.30}$$

and so on for every entry. Hence $\Gamma_5 Y \Gamma_5 = Y^\dagger$.

W and Y do not commute. K contains γ_μ, and $\gamma_\mu P_+ = P_-\gamma_\mu$, so a hop of K exchanges the two projectors of Y: $[Y, W] \neq 0$. This one fact makes the Möbius case different from Shamir's.

Consequence: $\Gamma_5 D \Gamma_5 \neq D^\dagger$. From the three rules above and $D = D_+ - D_- Y$,

$$\Gamma_5 D \Gamma_5 = \Gamma_5 D_+ \Gamma_5 - \Gamma_5 D_- \Gamma_5 \, \Gamma_5 Y \Gamma_5 = D_+^\dagger - D_-^\dagger Y^\dagger, \tag{1.31}$$

$$D^\dagger = D_+^\dagger - (D_- Y)^\dagger = D_+^\dagger - Y^\dagger D_-^\dagger. \tag{1.32}$$

The order of the last product differs. Subtracting,

$$\Gamma_5 D \Gamma_5 - D^\dagger = Y^\dagger D_-^\dagger - D_-^\dagger Y^\dagger \tag{1.33}$$

$$= Y^\dagger (cW^\dagger - 1) - (cW^\dagger - 1) Y^\dagger \tag{1.34}$$

$$= c\,(Y^\dagger W^\dagger - W^\dagger Y^\dagger) = c\,[Y^\dagger, W^\dagger], \tag{1.35}$$

which is of order c (zero for Shamir, 0.15 relative on a 12^4 stout configuration at $c = 0.5$).

The exact relation. D_+ and D_- are functions of W, so they commute. Factor D_- out to the left (formally):

$$D = D_- K_D, \qquad K_D = D_-^{-1} D_+ - Y. \tag{1.36}$$

K_D is Γ_5-Hermitian:

$$\Gamma_5 K_D \Gamma_5 = \Gamma_5 D_-^{-1} \Gamma_5 \, \Gamma_5 D_+ \Gamma_5 - \Gamma_5 Y \Gamma_5 \tag{1.37}$$

$$= (D_-^\dagger)^{-1} D_+^\dagger - Y^\dagger \tag{1.38}$$

$$= D_+^\dagger (D_-^\dagger)^{-1} - Y^\dagger \qquad (D_\pm^\dagger \text{ commute}) \tag{1.39}$$

$$= (D_-^{-1} D_+)^\dagger - Y^\dagger = K_D^\dagger. \tag{1.40}$$

So

$$D^\dagger = K_D^\dagger D_-^\dagger \tag{1.41}$$

$$= \Gamma_5 K_D \Gamma_5 \, D_-^\dagger \tag{1.42}$$

$$= \Gamma_5 D_-^{-1} D \, \Gamma_5 \, D_-^\dagger, \tag{1.43}$$

and multiplying by $D_- \Gamma_5$ from the left,

$$\boxed{D_- \, \Gamma_5 \, D^\dagger = D \, \Gamma_5 \, D_-^\dagger} \tag{1.44}$$

with no inverse left. D_- itself cannot be inverted in practice: $D_- = c(W - 1/c)$ is c times a Wilson operator of mass $-M_5 - 1/c = -3$, whose spectrum surrounds the origin (GCR stalls at $|r|^2/|b|^2 \approx 0.74$).

1.6 The site-diagonal part C and its inverse (note Sec. IV.C)

Y maps $P_+\psi$ to $P_+\psi$ and $P_-\psi$ to $P_-\psi$, so C splits into a block on $\psi^+ = P_+\psi$ and one on $\psi^- = P_-\psi$. On ψ^+, Y brings ψ^+_{s-1} to slice s with the factor -1, and across the wall the factor is $-(-m) = +m$:

$$(C\psi)^+_s = \begin{cases} d_+ \psi^+_0 + m\, d_- \psi^+_{L_s-1}, & s = 0, \\ d_+ \psi^+_s - d_- \psi^+_{s-1}, & s \geq 1. \end{cases} \tag{1.45}$$

To solve $C^+ x = r$, take the rows $s \geq 1$ first:

$$d_+ x_s - d_- x_{s-1} = r_s \quad \Longrightarrow \quad x_s = \frac{r_s + d_- x_{s-1}}{d_+}. \tag{1.46}$$

Each x_s is then affine in the unknown x_0. With $q = d_-/d_+$ we claim

$$x_s = p_s + q^s x_0, \qquad p_0 = 0, \qquad p_s = \frac{r_s + d_- p_{s-1}}{d_+}. \tag{1.47}$$

Check by induction:

$$\begin{aligned} x_s &= \frac{r_s + d_- x_{s-1}}{d_+} && (1.48)\\ &= \frac{r_s + d_-(p_{s-1} + q^{s-1}x_0)}{d_+} && (1.49)\\ &= \frac{r_s + d_- p_{s-1}}{d_+} + \frac{d_-}{d_+} q^{s-1} x_0 && (1.50)\\ &= p_s + q^s x_0. && (1.51) \end{aligned}$$

The remaining row $s = 0$ fixes x_0:

$$\begin{aligned} d_+ x_0 + m\, d_- x_{L_s-1} &= r_0 && (1.52)\\ d_+ x_0 + m\, d_-(p_{L_s-1} + q^{L_s-1} x_0) &= r_0 && (1.53)\\ (d_+ + m\, d_- q^{L_s-1}) x_0 &= r_0 - m\, d_- p_{L_s-1} && (1.54)\\ x_0 &= \frac{r_0 - m\, d_- p_{L_s-1}}{d_+ + m\, d_- q^{L_s-1}}. && (1.55) \end{aligned}$$

The block on ψ^- is the same with $s \to L_s - 1 - s$. So C^{-1} costs $O(L_s)$ per site, is exact, and does not depend on the gauge field. $C^\dagger$ has the transposed blocks and is inverted the same way from the other end.

1.7 The Pauli–Villars preconditioned operator (note Sec. IV.D)

Kanamori, Chen and Matsufuru multiply $D\psi = \eta$ by $(D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger$ from the left and set $\psi = C^{-1} y$:

$$\begin{aligned} D\psi &= \eta && (1.56)\\ DC^{-1} y &= \eta && (1.57)\\ (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1} y &= (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger \eta && (1.58)\\ A' y &= (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger \eta, \qquad A' \equiv (D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1})^\dagger D C^{-1}. && (1.59) \end{aligned}$$

At $m = 1$ it is Hermitian positive. Then $D = D_{\mathrm{PV}}$ and $C = C_{\mathrm{PV}}$, so with $X = D_{\mathrm{PV}} C_{\mathrm{PV}}^{-1}$,

$$A' = X^\dagger X, \qquad z^\dagger A' z = \|Xz\|^2 > 0 \;\; (z \neq 0,\; X \text{ invertible}). \tag{1.60}$$

At $m < 1$ it is a perturbation of that. m enters D only in the two corners of the matrix in s ($-mD_-P_+$ and $-mD_-P_-$), and C only in its corner $m\, d_-$:

$$D - D_{\mathrm{PV}} = (1 - m)\, D_- \left(P_+ \text{ in corner } (0, L_s - 1) + P_- \text{ in corner } (L_s - 1, 0)\right), \tag{1.61}$$

a term localized at the walls. So A' is $X^\dagger X$ plus a wall term; its eigenvalues stay in the right half-plane (note Figs. 2–3), which is what a Krylov method and a polynomial smoother need.

1.8 Neff's α (note Sec. IV.F)

Neff's operator is $D^{(\alpha)} = DQ$ with the site-local

$$Q_0 = P_- + \alpha P_+, \qquad Q_s = \alpha \ (0 < s < L_s - 1), \qquad Q_{L_s-1} = P_+ + \alpha P_-. \tag{1.62}$$

Q is Hermitian and Q^{-1} is the same with $\alpha \to 1/\alpha$. If $D^{(\alpha)}\psi^{(\alpha)} = \eta$, then

$$\eta = D^{(\alpha)}\psi^{(\alpha)} = DQ\psi^{(\alpha)} \quad \Longrightarrow \quad \psi = Q\psi^{(\alpha)} \text{ solves } D\psi = \eta. \tag{1.63}$$

Q leaves $P_-\psi_0$ and $P_+\psi_{L_s-1}$ alone, so the quark propagator is unchanged.

2 Even-odd forms (note Sec. V)

2.1 Blocks

Order the sites even first. K connects nearest neighbours, which have opposite parity, so in Eq. (1.21) the diagonal blocks come from C alone:

$$D = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}, \qquad D_{pq} = -\tfrac{1}{2}K_{pq}(b - cY) \quad (p \neq q). \tag{2.1}$$

C_e and C_o are the same matrix C on the even and the odd sites.

2.2 The Schur complement

Solve $D\psi = \eta$ block by block. The odd rows:

$$D_{oe}\psi_e + C_o\psi_o = \eta_o \tag{2.2}$$

$$\psi_o = C_o^{-1}(\eta_o - D_{oe}\psi_e). \tag{2.3}$$

Insert into the even rows:

$$C_e\psi_e + D_{eo}\psi_o = \eta_e \tag{2.4}$$

$$C_e\psi_e + D_{eo}C_o^{-1}(\eta_o - D_{oe}\psi_e) = \eta_e \tag{2.5}$$

$$(C_e - D_{eo}C_o^{-1}D_{oe})\psi_e = \eta_e - D_{eo}C_o^{-1}\eta_o \tag{2.6}$$

$$S\psi_e = \eta'_e, \qquad S \equiv C_e - D_{eo}C_o^{-1}D_{oe}, \quad \eta'_e \equiv \eta_e - D_{eo}C_o^{-1}\eta_o. \tag{2.7}$$

2.3 $S^{-1} = \pi D^{-1}\iota$

Let ι put an even-site field on the whole lattice (zero on the odd sites) and π take the even part:

$$\iota y = \begin{pmatrix} y \\ 0 \end{pmatrix}, \qquad \pi \begin{pmatrix} \psi_e \\ \psi_o \end{pmatrix} = \psi_e. \tag{2.8}$$

For $\eta = \iota\eta_e$ we have $\eta_o = 0$, so $\eta'_e = \eta_e$ and Eq. (2.7) reads $S\psi_e = \eta_e$. Hence

$$S^{-1}\eta_e = \psi_e = \pi D^{-1}\iota\eta_e \quad \Longrightarrow \quad S^{-1} = \pi D^{-1}\iota. \tag{2.9}$$

The same for $D^\dagger$: its block form is $\begin{pmatrix} C_e^\dagger & D_{oe}^\dagger \\ D_{eo}^\dagger & C_o^\dagger \end{pmatrix}$, and elimination gives $C_e^\dagger - D_{oe}^\dagger(C_o^\dagger)^{-1}D_{eo}^\dagger = S^\dagger$, so $S^{-\dagger} = \pi D^{-\dagger}\iota$.

2.4 The Pauli–Villars preconditioned forms

S is not suitable for a Krylov method: its spectrum surrounds the origin. Following Kanamori, Chen and Matsufuru we multiply from the left by the adjoint of the Pauli–Villars factor and divide the site part out. Two placements of C_e give two forms:

$$\tilde{A}_R = (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \, (S C_e^{-1}), \tag{2.10}$$

$$\tilde{A}_L = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger \, (C_e^{-1} S). \tag{2.11}$$

For $m = 1$ both become $X^\dagger X$, Hermitian positive; for $m < 1$ they are perturbations of it, with all eigenvalues in the right half-plane.

Right form, step by step. Start from $S\psi_e = \eta'_e$ and substitute $\psi_e = C_e^{-1} y$:

$$S C_e^{-1} y = \eta'_e \tag{2.12}$$

$$(S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger S C_e^{-1} y = (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \eta'_e \tag{2.13}$$

$$\tilde{A}_R \, y = \rho_R, \qquad \rho_R \equiv (S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \eta'_e, \tag{2.14}$$

$$\psi_e = C_e^{-1} y, \qquad \psi_o = C_o^{-1}(\eta_o - D_{oe}\psi_e). \tag{2.15}$$

Left form with α in D only. For DQ the blocks are $C_e Q_e$, $D_{eo} Q_o$, $D_{oe} Q_e$, $C_o Q_o$, and the Schur complement is

$$S^{(\alpha)} = C_e Q_e - D_{eo} Q_o (C_o Q_o)^{-1} D_{oe} Q_e \tag{2.16}$$

$$= C_e Q_e - D_{eo} Q_o Q_o^{-1} C_o^{-1} D_{oe} Q_e \tag{2.17}$$

$$= (C_e - D_{eo} C_o^{-1} D_{oe}) Q_e \tag{2.18}$$

$$= S Q_e. \tag{2.19}$$

So the left Schur factor of DQ is

$$K^{(\alpha)} = (C_e Q_e)^{-1} S Q_e = Q_e^{-1} C_e^{-1} S Q_e = Q_e^{-1} K Q_e, \qquad K \equiv C_e^{-1} S, \tag{2.20}$$

and the left form becomes $\tilde{A}_L^{(\alpha)} = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger Q_e^{-1} (C_e^{-1} S) Q_e$. In the right form the two Q_e cancel: $S Q_e (C_e Q_e)^{-1} = S C_e^{-1}$, so $\tilde{A}_R$ does not see α.

3 The factorized coarse-grid correction and the prolongation (note Sec. VI)

3.1 The factorized coarse correction

The inverse of the right form. Invert the product one factor at a time:

$$\tilde{A}_R^{-1} = \left[(S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger (S C_e^{-1}) \right]^{-1} \tag{3.1}$$

$$= (S C_e^{-1})^{-1} \left[(S_{\mathrm{PV}} C_{\mathrm{PV},e}^{-1})^\dagger \right]^{-1} \tag{3.2}$$

$$= C_e S^{-1} \left[(C_{\mathrm{PV},e}^{-1})^\dagger S_{\mathrm{PV}}^\dagger \right]^{-1} \tag{3.3}$$

$$= C_e S^{-1} \, (S_{\mathrm{PV}}^\dagger)^{-1} C_{\mathrm{PV},e}^\dagger \tag{3.4}$$

$$= C_e \, \pi D^{-1} \iota \;\; \pi (D_{\mathrm{PV}}^\dagger)^{-1} \iota \;\; C_{\mathrm{PV},e}^\dagger, \tag{3.5}$$

where Eq. (2.9) was used twice. The inverse is a product of the inverses of two *one-hop* operators on the whole lattice, D and $D_{\mathrm{PV}}^\dagger$, and site-local factors.

Galerkin replacement. With a prolongation $\mathcal{P}$ (coarse to whole lattice), replace each inverse by its Galerkin approximation:

$$X^{-1} \approx \mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger. \tag{3.6}$$

Both coarse operators $\mathcal{D}_c = \mathcal{P}^\dagger D \mathcal{P}$ and $\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}$ are nearest-neighbour on the coarse lattice, because D is one hop and every aggregate contains the whole fifth dimension.

Why the Galerkin approximation is exact on range $\mathcal{P}$**.** For any X with $\mathcal{P}^\dagger X \mathcal{P}$ invertible and any coarse vector c,

$$\mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}\mathcal{P}^\dagger X\,(\mathcal{P}c) = \mathcal{P}(\mathcal{P}^\dagger X \mathcal{P})^{-1}(\mathcal{P}^\dagger X \mathcal{P})c = \mathcal{P}c. \tag{3.7}$$

So the approximation of X^{-1} inverts X exactly on vectors $X\mathcal{P}c$. This is used in the exactness proof below.

The size of the coarse stencil. An operator that reaches k hops on the fine lattice connects a fine site to sites at most k hops away. With blocks at least two sites wide, two coarse sites can then be coupled only if their coordinates differ by at most 1 in at most k of the four directions. Counting the coarse neighbours that differ in exactly j directions ($\binom{4}{j}$ choices of directions, 2^j signs):

$$k = 1: \quad 1 + \tbinom{4}{1}2 = 1 + 8 = 9, \tag{3.8}$$

$$k = 2: \quad 1 + 8 + \tbinom{4}{2}2^2 = 1 + 8 + 24 = 33, \tag{3.9}$$

$$k \geq 4: \quad \sum_{j=0}^{4} \tbinom{4}{j}2^j = 3^4 = 81. \tag{3.10}$$

D and D_{PV} are one hop: 9 points each. $\tilde{A}_R$ reaches four hops (S is two hops, and $\tilde{A}_R$ contains S and $S_{\mathrm{PV}}^\dagger$): 81 points. That is why the correction factorizes instead of coarsening $\tilde{A}_R$ directly.

The correction of a residual r (Eq. (3.5) read from the right) is

$$e = O_L\,\pi\mathcal{P}\,\mathcal{D}_c^{-1}\mathcal{P}^\dagger\iota\,O_M\,\pi\mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1}\mathcal{P}^\dagger\iota\,O_R\,r \tag{3.11}$$

with the site-local factors

form	O_R	O_M	O_L
right, any α	$C_{\mathrm{PV},e}^\dagger$	1	C_e
left, α in D only	1	$C_e Q_e C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}
left, α in D and D_{PV}	Q_e^{-1}	$C_e Q_e^2 C_{\mathrm{PV},e}^\dagger$	Q_e^{-1}

In words: (i) put $O_R r$ on the whole lattice, restrict, solve with the adjoint Pauli–Villars coarse operator, prolong; (ii) take the even part, multiply by O_M, put it back on the whole lattice, restrict, solve with the quark coarse operator, prolong; (iii) take the even part and multiply by O_L (Fig. 1). The coarse operators are always those of the plain D and D_{PV}, so one setup serves both forms and every α.

The left form with α, step by step. For α in D only,

$$\tilde{A}_L^{(\alpha)\,-1} = \left[(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\, Q_e^{-1}C_e^{-1}SQ_e\right]^{-1} \tag{3.12}$$

$$= \left(Q_e^{-1}C_e^{-1}SQ_e\right)^{-1}\left[(C_{\mathrm{PV},e}^{-1}S_{\mathrm{PV}})^\dagger\right]^{-1} \tag{3.13}$$

$$= Q_e^{-1}S^{-1}C_eQ_e\left[S_{\mathrm{PV}}^\dagger(C_{\mathrm{PV},e}^{-1})^\dagger\right]^{-1} \tag{3.14}$$

$$= Q_e^{-1}S^{-1}C_eQ_e\,C_{\mathrm{PV},e}^\dagger\,(S_{\mathrm{PV}}^\dagger)^{-1} \tag{3.15}$$

$$= \underbrace{Q_e^{-1}}_{O_L}\,\pi D^{-1}\iota\,\underbrace{C_eQ_eC_{\mathrm{PV},e}^\dagger}_{O_M}\,\pi(D_{\mathrm{PV}}^\dagger)^{-1}\iota\,\underbrace{1}_{O_R}. \tag{3.16}$$

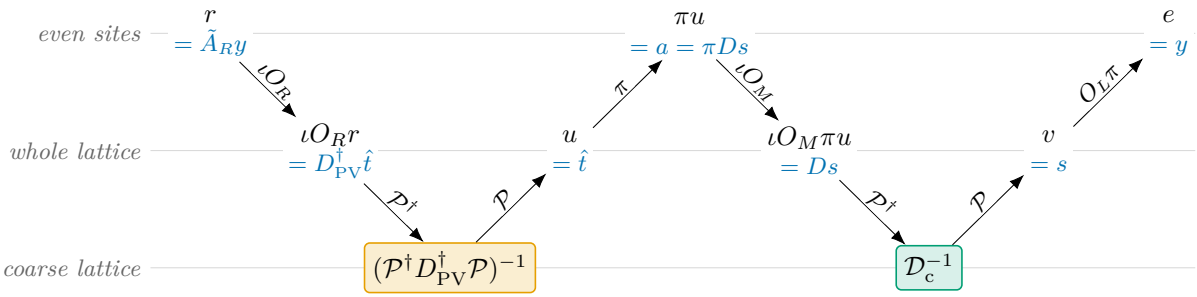


Figure 1: One factorized coarse correction, read from the left, on its three lattices (the Pauli–Villars coarse solve in orange, the quark one in green): $u = \mathcal{P}(\mathcal{P}^{\dagger} D_{\mathrm{PV}}^{\dagger} \mathcal{P})^{-1} \mathcal{P}^{\dagger} \iota O_R r$ is the result of the first coarse solve, its even part times O_M is the right-hand side of the second, whose result is $v = \mathcal{P} \mathcal{D}_{\mathrm{c}}^{-1} \mathcal{P}^{\dagger} \iota O_M \pi u$, and $e = O_L \pi v$. In blue, after "=", the values for $r = \tilde{A}_R y$ in the right form with exact coarse solves and $s, \hat{t} \in \operatorname{range} \mathcal{P}$ (Sec. 3.4; $O_R = C_{\mathrm{PV},e}^{\dagger}$, $O_M = 1$, $O_L = C_e$): the first solve returns the lift $\hat{t}$, whose even part is $a = \pi D s$, the second returns the lift s, and the correction returns y.

Unlike the right form, the Galerkin factor of $D_{\mathrm{PV}}^{\dagger}$ acts on the plain residual, and $C_{\mathrm{PV},e}^{\dagger}$ lands between the two coarse solves, in O_M.

3.2 Aggregation and the chirality split

The fine lattice is divided into 4^4 blocks, each with the whole fifth dimension; each block is one coarse site. Every test vector v is split by $\Gamma_5 = \gamma_5 R_5$, $(R_5 \psi)_s = \psi_{L_s - 1 - s}$:

$$v_{\pm} = \tfrac{1}{2}(1 \pm \Gamma_5) v, \qquad \Gamma_5 v_{\pm} = \pm v_{\pm}. \tag{3.17}$$

With n_v test vectors a coarse site has $2n_v$ components, the + parts first. Each column of $\mathcal{P}$ is then an eigenvector of Γ_5, so with $\Sigma = \operatorname{diag}(\mathbf{1}_{n_v}, -\mathbf{1}_{n_v})$ per coarse site

$$\Gamma_5 \mathcal{P} = \mathcal{P} \Sigma. \tag{3.18}$$

Half the storage. Write the condition $\Gamma_5 v_{\pm} = \pm v_{\pm}$ out slice by slice:

$$(\Gamma_5 v_{\pm})_s = \gamma_5 \, (R_5 v_{\pm})_s = \gamma_5 \, v_{\pm}(L_s - 1 - s) \tag{3.19}$$

$$\pm v_{\pm}(s) = \gamma_5 \, v_{\pm}(L_s - 1 - s) \tag{3.20}$$

$$v_{\pm}(L_s - 1 - s) = \pm \gamma_5 \, v_{\pm}(s). \tag{3.21}$$

So each part is fixed by its slices $s < L_s/2$; $\mathcal{P}$ is stored only there ("folded"), in half precision. Restriction and prolongation unfold on the fly: for the restriction of a fine field f,

$$(\mathcal{P}^{\dagger} f)_{X,k} = \sum_{x \in X} \sum_{s < L_s/2} \mathcal{P}^{*}_{x,s;k} \, (f_s \pm \gamma_5 f_{L_s - 1 - s}) \tag{3.22}$$

(the sign that of the chirality of column k), which is what `fold_restrict_tiled_kernel` computes.

3.3 Orthonormalization per block: CholeskyQR2

In each block and chirality the columns $w_1, \ldots, w_n$ ($n = 2\times$ the number of filtered vectors, see below) are made orthonormal. Collect them as the columns of W and write the QR decomposition

with a positive diagonal of R:

$$W = QR \tag{3.23}$$
$$W^\dagger W = R^\dagger Q^\dagger QR = R^\dagger R \tag{3.24}$$
$$G \equiv W^\dagger W = R^\dagger R \quad \text{(Cholesky)} \tag{3.25}$$
$$Q = WR^{-1}. \tag{3.26}$$

So one Gram matrix ($n \times n$), one Cholesky factor and one triangular solve give Q; the factorization is unique, so this Q equals what modified Gram–Schmidt produces. In floating point one pass loses orthogonality like the square of the condition number of W, so the pass is done twice (CholeskyQR2): the second G is close to the identity and repairs the first. Each pass reads W twice, against one sweep per pair of columns for Gram–Schmidt (800 sweeps for $n = 40$). Where a pivot of the first pass falls below 10^{-12} of its diagonal (nearly dependent columns), the code falls back to Gram–Schmidt for those blocks.

3.4 The basis $\{s, \hat{t}\}$ and why the correction is exact

The setup produces vectors y_j on the even sites (next section). Each is lifted to the whole lattice in two ways, one for each coarse solve:

$$s = (C_e^{-1}y,\ -C_o^{-1}D_{oe}C_e^{-1}y), \tag{3.27}$$
$$\hat{t} = (a,\ -(C_{\mathrm{PV},o}^\dagger)^{-1}D_{\mathrm{PV},eo}^\dagger a), \qquad a = \pi Ds. \tag{3.28}$$

Ds vanishes on the odd sites.

$$Ds = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix} \begin{pmatrix} C_e^{-1}y \\ -C_o^{-1}D_{oe}C_e^{-1}y \end{pmatrix} \tag{3.29}$$
$$= \begin{pmatrix} C_eC_e^{-1}y - D_{eo}C_o^{-1}D_{oe}C_e^{-1}y \\ D_{oe}C_e^{-1}y - C_oC_o^{-1}D_{oe}C_e^{-1}y \end{pmatrix} \tag{3.30}$$
$$= \begin{pmatrix} (C_e - D_{eo}C_o^{-1}D_{oe})C_e^{-1}y \\ 0 \end{pmatrix} \tag{3.31}$$
$$= \iota\, SC_e^{-1}y. \tag{3.32}$$

In the same way $D_{\mathrm{PV}}^\dagger \hat{t} = \iota S_{\mathrm{PV}}^\dagger a$.

Exactness of the right form. Let $s, \hat{t} \in \operatorname{range}\mathcal{P}$ and solve both coarse systems exactly. The Galerkin map $\mathcal{P}(\mathcal{P}^\dagger X\mathcal{P})^{-1}\mathcal{P}^\dagger X$ is the identity on $\operatorname{range}\mathcal{P}$. Follow $r = \tilde{A}_R y$ through the correction ($O_R = C_{\mathrm{PV},e}^\dagger$, $O_M = 1$, $O_L = C_e$):

$$O_R r = C_{\mathrm{PV},e}^\dagger (S_{\mathrm{PV}}C_{\mathrm{PV},e}^{-1})^\dagger SC_e^{-1}y \tag{3.33}$$
$$= C_{\mathrm{PV},e}^\dagger (C_{\mathrm{PV},e}^{-1})^\dagger S_{\mathrm{PV}}^\dagger SC_e^{-1}y \tag{3.34}$$
$$= S_{\mathrm{PV}}^\dagger\, a, \qquad (a = SC_e^{-1}y = \pi Ds) \tag{3.35}$$
$$\iota O_R r = \iota S_{\mathrm{PV}}^\dagger a = D_{\mathrm{PV}}^\dagger \hat{t} \tag{3.36}$$
$$\mathcal{P}(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P})^{-1}\mathcal{P}^\dagger\, D_{\mathrm{PV}}^\dagger \hat{t} = \hat{t} \qquad (\hat{t} \in \operatorname{range}\mathcal{P}) \tag{3.37}$$
$$\pi\hat{t} = a = \pi Ds \tag{3.38}$$
$$\iota O_M \pi\hat{t} = \iota\pi Ds = Ds \qquad (Ds \text{ is zero on the odd sites}) \tag{3.39}$$
$$\mathcal{P}\mathcal{D}_{\mathrm{c}}^{-1}\mathcal{P}^\dagger Ds = s \qquad (s \in \operatorname{range}\mathcal{P}) \tag{3.40}$$
$$e = O_L \pi s = C_eC_e^{-1}y = y. \tag{3.41}$$

So the correction inverts $\tilde{A}_R$ exactly on the span of the y_j. In practice $\mathcal{P}$ is in half precision and the coarse solves are inexact (relative residual 10^{-1}), and the left form is not exact at all; the cycle converges nonetheless.

Why the left form is not exact. Repeat the argument for $\tilde{A}_L$ with $\alpha = 1$ and $r = \tilde{A}_L y'$, $y' = C_e^{-1} y$ ($O_R = 1$, $O_M = C_e C_{\mathrm{PV},e}^\dagger$, $O_L = 1$):

$$r = (C_{\mathrm{PV},e}^{-1} S_{\mathrm{PV}})^\dagger C_e^{-1} S\, C_e^{-1} y = S_{\mathrm{PV}}^\dagger (C_{\mathrm{PV},e}^{-1})^\dagger C_e^{-1} a, \qquad a = S C_e^{-1} y, \tag{3.42}$$

$$\iota O_R r = \iota S_{\mathrm{PV}}^\dagger u, \qquad u \equiv (C_{\mathrm{PV},e}^\dagger)^{-1} C_e^{-1} a. \tag{3.43}$$

The first solve returns the Pauli–Villars lift of u, not of a. u differs from a by the site-local matrix $(C_{\mathrm{PV},e}^\dagger)^{-1} C_e^{-1}$, which mixes the slices s and the spin, so its lift is in general not in the span of the $\hat{t}_j$ made from a. The correction is then only approximate. In practice the left form converges within a few iterations of the right form (note Sec. XII), so the cycle does not need the exactness.

3.5 The coarse operators by probing

$\mathcal{D}_{\mathrm{c}} = \mathcal{P}^\dagger D \mathcal{P}$ has on each coarse site X a $2n_v \times 2n_v$ diagonal block $A(X)$ and one link block $B_\mu^\pm(X)$ per direction: nine blocks. They are found column by column. For component k, put a unit vector ϵ_k on *every* coarse site and prolong: $\phi = \mathcal{P}\epsilon_k$. Split $D = C + H$ (H the hops) and H by leg (direction and sign), and split each leg into the part that stays inside its block and the part that crosses to the neighbour block:

$$D\phi = C\phi + \sum_{\text{legs } l} H_l \phi \tag{3.44}$$

$$= \underbrace{C\phi + \sum_l H_l^{\mathrm{in}} \phi}_{\text{stays in } X} + \sum_l \underbrace{H_l^{\mathrm{cross}} \phi}_{\text{comes from } X \pm \hat{\mu}} . \tag{3.45}$$

Restricting, $\big(\mathcal{P}^\dagger (C\phi + \sum_l H_l^{\mathrm{in}} \phi)\big)_X$ is column k of $A(X)$, and $\big(\mathcal{P}^\dagger H_l^{\mathrm{cross}} \phi\big)_X$ is column k of the link block of leg l, since every coarse site carries the same unit vector and a crossing hop brings exactly the neighbour's contribution. Doing this for all $2n_v$ components fills all nine blocks (`coarse_blocks`, `hop_split`, `restrict_leg`). Several components are probed at a time, as many as a quarter of the free device memory allows (four on 24^4), so that $\mathcal{P}$ is read once for them.

4 The coarse solvers (note Sec. VII)

Quark factor. $\mathcal{D}_{\mathrm{c}} c = g$ by BiCGStab on its even-odd Schur complement, preconditioned by the inverse diagonal blocks. With $\mathcal{D}_{\mathrm{c}} = \begin{pmatrix} A_{ee} & B_{eo} \\ B_{oe} & A_{oo} \end{pmatrix}$ on the coarse lattice, exactly as in Eq. (2.7),

$$(A_{ee} - B_{eo} A_{oo}^{-1} B_{oe}) c_e = g_e - B_{eo} A_{oo}^{-1} g_o, \tag{4.1}$$

$$c_o = A_{oo}^{-1} (g_o - B_{oe} c_e). \tag{4.2}$$

Pauli–Villars factor. $(\mathcal{P}^\dagger D_{\mathrm{PV}}^\dagger \mathcal{P}) w = g$. With $\Gamma_5 \mathcal{P} = \mathcal{P}\Sigma$,

$$\mathcal{P}^\dagger (\Gamma_5 D_{\mathrm{PV}} \Gamma_5) \mathcal{P} = \mathcal{P}^\dagger \Gamma_5\, D_{\mathrm{PV}}\, \Gamma_5 \mathcal{P} \tag{4.3}$$

$$= (\Gamma_5 \mathcal{P})^\dagger\, D_{\mathrm{PV}}\, (\Gamma_5 \mathcal{P}) \tag{4.4}$$

$$= \Sigma \mathcal{P}^\dagger D_{\mathrm{PV}} \mathcal{P} \Sigma = \Sigma \mathcal{D}_{\mathrm{c}}^{\mathrm{PV}} \Sigma. \tag{4.5}$$

If $\Gamma_5 D_{\rm PV}\Gamma_5 = D_{\rm PV}^\dagger$ (true for $c = 0$, off by $O(c)$ otherwise, Sec. 8), the adjoint coarse factor is $\Sigma\mathcal{D}_{\rm c}^{\rm PV}\Sigma$, and

$$\Sigma\mathcal{D}_{\rm c}^{\rm PV}\Sigma\, w = g \tag{4.6}$$

$$\mathcal{D}_{\rm c}^{\rm PV}\Sigma\, w = \Sigma g \qquad (\Sigma^2 = 1) \tag{4.7}$$

$$\mathcal{H}_c\, w = \Sigma g, \qquad \mathcal{H}_c \equiv \mathcal{D}_{\rm c}^{\rm PV}\Sigma, \tag{4.8}$$

$\mathcal{H}_c$ Hermitian (indefinite) for $c = 0$, solved by MINRES with the absolute value of the diagonal blocks as a positive preconditioner. Both factors stop at $|r|^2/|b|^2 < 10^{-2}$.

Why $\mathcal{H}_c$ is Hermitian when $\Gamma_5 D_{\rm PV}\Gamma_5 = D_{\rm PV}^\dagger$. Then $(\mathcal{D}_{\rm c}^{\rm PV})^\dagger = \Sigma\mathcal{D}_{\rm c}^{\rm PV}\Sigma$, and with $\Sigma^\dagger = \Sigma$, $\Sigma^2 = 1$,

$$\mathcal{H}_c^\dagger = (\mathcal{D}_{\rm c}^{\rm PV}\Sigma)^\dagger = \Sigma(\mathcal{D}_{\rm c}^{\rm PV})^\dagger = \Sigma\Sigma\mathcal{D}_{\rm c}^{\rm PV}\Sigma = \mathcal{D}_{\rm c}^{\rm PV}\Sigma = \mathcal{H}_c. \tag{4.9}$$

For $c \neq 0$ this holds only approximately (the non-Hermitian part is a few percent, note Sec. XII.H), and MINRES is used anyway.

The even-odd form of $\mathcal{H}_c$. Σ is site-local, so with the coarse blocks of $\mathcal{D}_{\rm c}^{\rm PV}$ the blocks of $\mathcal{H}_c = \mathcal{D}_{\rm c}^{\rm PV}\Sigma$ are $A_{ee}\Sigma_e$, $B_{eo}\Sigma_o$, $B_{oe}\Sigma_e$, $A_{oo}\Sigma_o$. Its Schur complement:

$$A_{ee}\Sigma_e - B_{eo}\Sigma_o(A_{oo}\Sigma_o)^{-1}B_{oe}\Sigma_e = A_{ee}\Sigma_e - B_{eo}\Sigma_o\Sigma_o^{-1}A_{oo}^{-1}B_{oe}\Sigma_e \tag{4.10}$$

$$= (A_{ee} - B_{eo}A_{oo}^{-1}B_{oe})\Sigma_e = S_c\Sigma_e, \tag{4.11}$$

the Schur complement S_c of $\mathcal{D}_{\rm c}^{\rm PV}$ times Σ_e. So MINRES runs on the even coarse sites with $u_e = \Sigma w_e$, and the odd sites follow as for the quark factor.

The preconditioner of MINRES. MINRES needs a Hermitian positive preconditioner. On each coarse site take the Hermitian part of the diagonal block, $\frac{1}{2}(\mathcal{H}_{xx} + \mathcal{H}_{xx}^\dagger) = V\Lambda V^\dagger$, and use

$$|\mathcal{H}_{xx}|^{-1} = V|\Lambda|^{-1}V^\dagger, \qquad z^\dagger|\mathcal{H}_{xx}|^{-1}z = \sum_k \frac{|(V^\dagger z)_k|^2}{|\lambda_k|} > 0, \tag{4.12}$$

Hermitian and positive however indefinite $\mathcal{H}_{xx}$ is.

Levels below. At small masses the coarse solves themselves become expensive (on 24^4 at $m = 0.001$ about 2000 applications of $\mathcal{D}_{\rm c}$ per correction). The coarse sites are then grouped into super-blocks (2, then 3), local vectors are formed from $\mathcal{P}^\dagger s_j$, $\mathcal{P}^\dagger \hat{t}_j$, and a V-cycle preconditions the coarse Krylov solve: with Q_ℓ the prolongation into level $\ell - 1$ and B_ℓ its diagonal blocks,

$$z = Q_{\ell+1}M_{\ell+1}Q_{\ell+1}^\dagger r, \qquad M_\ell r = z + B_\ell^{-1}(r - \mathcal{D}_\ell z), \tag{4.13}$$

the last level inverted densely. This brings the coarse cost down to about 50 applications per correction.

5 Setup (note Sec. VIII)

The setup produces the vectors y_j, then $\{s_j, \hat{t}_j\}$, $\mathcal{P}$, the coarse operators and the levels. It always uses the right form.

5.1 Chebyshev-filtered subspace iteration

The y_j should approximate the low right singular vectors of $\tilde{A}_R$, i.e. the low eigenvectors of $H_n = \tilde{A}_R^\dagger \tilde{A}_R$. With $\ell(x) = (2x - (\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}}))/(\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}})$, the polynomial $T_d(\ell(x))$ is bounded by 1 on $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ and grows fast below λ_{lo}. One round on a block X of m vectors:

$$X \leftarrow T_d(\ell(H_n))X \qquad \text{(three-term recurrence, } d \text{ applications of } H_n) \tag{5.1}$$

$$X \leftarrow \text{orthonormalize } X \tag{5.2}$$

$$G = X^\dagger H_n X = V \Theta V^\dagger, \quad X \leftarrow XV \qquad \text{(Rayleigh–Ritz)} \tag{5.3}$$

$$\lambda_{\mathrm{lo}} \leftarrow \theta_{\max}. \tag{5.4}$$

How much the filter amplifies, and the degree. For $|t| \geq 1$, $|T_d(t)| = \cosh(d \operatorname{arccosh}|t|)$. A mode with x just below λ_{lo} has $\ell(x) = -(1+\delta)$ with small $\delta > 0$, and

$$\operatorname{arccosh}(1+\delta) = \ln\Big(1 + \delta + \sqrt{2\delta + \delta^2}\Big) \approx \sqrt{2\delta}, \tag{5.5}$$

$$|T_d(\ell(x))| \approx \cosh(d\sqrt{2\delta}). \tag{5.6}$$

The lowest modes have $x \approx 0$, where

$$\ell(0) = -\frac{\lambda_{\mathrm{hi}} + \lambda_{\mathrm{lo}}}{\lambda_{\mathrm{hi}} - \lambda_{\mathrm{lo}}} \approx -\Big(1 + 2\frac{\lambda_{\mathrm{lo}}}{\lambda_{\mathrm{hi}}}\Big), \qquad \delta \approx 2\frac{\lambda_{\mathrm{lo}}}{\lambda_{\mathrm{hi}}}, \qquad |T_d(\ell(0))| \approx \cosh\Big(2d\sqrt{\lambda_{\mathrm{lo}}/\lambda_{\mathrm{hi}}}\Big), \tag{5.7}$$

while every mode inside $[\lambda_{\mathrm{lo}}, \lambda_{\mathrm{hi}}]$ stays below 1. With the code's degree $d = 1.5\sqrt{\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}}$ the lowest modes gain a factor $\cosh 3 \approx 10$ per round over the damped ones: the degree is chosen for a fixed amplification per round.

In the code the degree is $d = 1.5\sqrt{\lambda_{\mathrm{hi}}/\lambda_{\mathrm{lo}}}$, between 40 and 1000; the first round uses $\lambda_{\mathrm{lo}} = \lambda_{\mathrm{hi}}/1000$. *Observation* ($24^4$, $m = 0.001$, 20 vectors): after the first round (degree 48) the largest Ritz value is still $0.34\,\lambda_{\mathrm{hi}}$, so the second round has degree 40 and damps only the top two thirds of the spectrum. The filter is weak; the bootstrap does most of the work.

5.2 Bootstrap

Each round: build $\mathcal{P}$ and the cycle $\mathcal{M}$ from the current y; replace every y_j by four steps of FGMRES on $\tilde{A}_R x = y_j$ from $x = 0$ preconditioned by $\mathcal{M}$; orthonormalize y through its Gram matrix. Why it works: write y_j in eigenvectors of the preconditioned operator. The cycle already handles the components inside its coarse space, and FGMRES with it removes them first; what remains, and is therefore amplified relative to the rest, is what the coarse space misses. In HMC the levels below are used from the second round on (in the first, on the filtered vectors, the coarse solves were as fast without them).

5.3 Cost and memory

On 24^4 ($L_s = 8$, 20 filtered vectors, $m = 0.001$, A30): the filter about 35 s and the bootstrap about 77 s, the setup in all about 113 s. A bootstrap round makes its correction ($\mathcal{P}$ 1.8 s, the two coarse operators about 4.4 s each, the inverse blocks 0.8 s, the top of the smoother 1.3 s; the levels 7 s more in the second round) and then improves the 20 vectors (about 20 s in the first round, 26 s in the second). $\mathcal{P}$ (40 columns, half precision, whole lattice) is the largest object, about 4.7 GB.

6 The Chebyshev smoother and the cycle (note Sec. IX)

One cycle $y = \mathcal{M}b$ is

$$y \leftarrow p(\tilde{A})\, b \qquad \text{(pre-smoothing)} \tag{6.1}$$
$$r \leftarrow b - \tilde{A}y \tag{6.2}$$
$$y \leftarrow y + e(r) \qquad \text{(coarse correction)} \tag{6.3}$$
$$r \leftarrow b - \tilde{A}y \tag{6.4}$$
$$y \leftarrow y + p(\tilde{A})\, r \qquad \text{(post-smoothing).} \tag{6.5}$$

Cost of one cycle. Count the applications of $\tilde{A}$: the pre-smoothing (ν Chebyshev steps from $x = 0$) costs $\nu - 1$, the residual after it 1, the residual after the correction 1, the post-smoothing $\nu - 1$:

$$(\nu - 1) + 1 + 1 + (\nu - 1) = 2\nu, \tag{6.6}$$

plus the two coarse solves. One application of $\tilde{A}$ is two Schur factors, each two hops: $\nu = 8$ means 16 applications, 64 hops, per cycle. This is why one multigrid iteration costs as much as about 24 iterations of CG on $K^\dagger K$ (two hops each with K and $K^\dagger$).

6.1 The Chebyshev smoother, derived

The smoother is ν steps of Chebyshev iteration on $\tilde{A}x = b$ from $x = 0$. Choose the interval $[\lambda_{\rm lo}, \lambda_{\rm hi}]$ that it damps ($\lambda_{\rm hi}$ slightly above the spectral radius from a power iteration, $\lambda_{\rm lo} = \lambda_{\rm hi}/\kappa_s$) and set

$$\theta = \tfrac{1}{2}(\lambda_{\rm hi} + \lambda_{\rm lo}), \qquad \delta = \tfrac{1}{2}(\lambda_{\rm hi} - \lambda_{\rm lo}), \qquad \sigma = \theta/\delta. \tag{6.7}$$

The residual after k steps is $r_k = R_k(\tilde{A})r_0$ with the scaled Chebyshev polynomial

$$R_k(\lambda) = \frac{T_k\big((\theta - \lambda)/\delta\big)}{T_k(\sigma)}, \qquad R_k(0) = 1, \tag{6.8}$$

which is at most $1/T_k(\sigma)$ in modulus on the interval. Write $t = (\theta - \lambda)/\delta$ and use $T_{k+1}(t) = 2tT_k(t) - T_{k-1}(t)$:

$$T_{k+1}(\sigma)R_{k+1} = 2t\,T_k(\sigma)R_k - T_{k-1}(\sigma)R_{k-1} \tag{6.9}$$
$$R_{k+1} = \rho_k\big[2t\,R_k - \rho_{k-1}R_{k-1}\big], \qquad \rho_k \equiv \frac{T_k(\sigma)}{T_{k+1}(\sigma)}. \tag{6.10}$$

The same recurrence at $t = \sigma$ gives the ρ's:

$$T_{k+1}(\sigma) = 2\sigma T_k(\sigma) - T_{k-1}(\sigma) \tag{6.11}$$
$$\frac{1}{\rho_k} = 2\sigma - \rho_{k-1}, \qquad \rho_0 = \frac{T_0(\sigma)}{T_1(\sigma)} = \frac{1}{\sigma}. \tag{6.12}$$

Now split $2t = 2\sigma - 2\lambda/\delta$ and use $2\sigma = 1/\rho_k + \rho_{k-1}$:

$$R_{k+1} = \rho_k\Big[(2\sigma - \tfrac{2\lambda}{\delta})R_k - \rho_{k-1}R_{k-1}\Big] \tag{6.13}$$
$$= \rho_k\Big[(\tfrac{1}{\rho_k} + \rho_{k-1})R_k - \tfrac{2\lambda}{\delta}R_k - \rho_{k-1}R_{k-1}\Big] \tag{6.14}$$
$$= R_k + \rho_k\rho_{k-1}(R_k - R_{k-1}) - \frac{2\rho_k}{\delta}\lambda R_k. \tag{6.15}$$

Apply to r_0, with $\lambda \to \tilde{A}$:

$$r_{k+1} - r_k = \rho_k\rho_{k-1}(r_k - r_{k-1}) - \frac{2\rho_k}{\delta}\tilde{A}r_k. \tag{6.16}$$

Since $r = b - \tilde{A}x$, a step $x_{k+1} = x_k + d_k$ changes the residual by $-\tilde{A}d_k$. Matching the two sides,

$$d_k = \rho_k \rho_{k-1}\, d_{k-1} + \frac{2\rho_k}{\delta}\, r_k. \tag{6.17}$$

The first step: $R_1 = T_1(t)/T_1(\sigma) = t/\sigma = 1 - \lambda/\theta$, so $x_1 = b/\theta$ and $d_0 = r_0/\theta$. These are the updates of the code (`BlasChebyshevStep`): ν steps cost $\nu - 1$ applications of $\tilde{A}$, and the polynomial is fixed, so the smoother is a linear map. The runs use $\nu = 8$, $\kappa_s = 10$ with stout links ($M_5 = 1$) and $\nu = 12$, $\kappa_s = 30$ without ($M_5 = 1.8$).

7 The outer solver: flexible GCR (note Sec. X)

The outer iteration solves $\tilde{A}y = \rho$ ($\tilde{A} = \tilde{A}_R$ or $\tilde{A}_L$) with a right preconditioner $\mathcal{M}$, the multigrid cycle. Because $\mathcal{M}$ contains Krylov solves stopped by a residual test, it changes from one application to the next, so the method must be flexible: it keeps the preconditioned directions z_i themselves.

One step of GCR(k), from the residual $r_i = \rho - \tilde{A}y_i$:

$$z_i = \mathcal{M} r_i \qquad \text{(one cycle)} \tag{7.1}$$

$$q_i = \tilde{A} z_i \qquad \text{(one application of } \tilde{A}) \tag{7.2}$$

$$q_i \leftarrow q_i - \sum_j \frac{(q_j, q_i)}{(q_j, q_j)} q_j, \quad z_i \leftarrow z_i - \sum_j \frac{(q_j, q_i)}{(q_j, q_j)} z_j \qquad \text{(the last } k \text{ kept pairs)} \tag{7.3}$$

$$\alpha_i = \frac{(q_i, r_i)}{(q_i, q_i)} \tag{7.4}$$

$$y_{i+1} = y_i + \alpha_i z_i, \qquad r_{i+1} = r_i - \alpha_i q_i. \tag{7.5}$$

The coefficient α_i minimizes $\|r_i - \alpha q_i\|$, and the orthogonalization keeps $\tilde{A}z_i = q_i$ true after the update, so $r_{i+1} = \rho - \tilde{A}y_{i+1}$ exactly. With $k = 2$ (GCR(2)) the code keeps $k + 1 = 3$ pairs (z, q) and the residual, seven fields on the even sites, at the price of no optimality over the whole Krylov space.

8 Solving with $D^\dagger$ (note Sec. XI)

HMC needs $(D^\dagger D)^{-1}$, so it solves with $D^\dagger$ as often as with D. A second setup for $D^\dagger$ would double the cost. The tool that avoids it is an exact relation between D and $D^\dagger$.

The Γ_5-transformation and the exact relation it rests on were derived in Sec. 1.5.

8.1 $\Gamma_5 D^{-1} \Gamma_5$ as a preconditioner of $D^\dagger$ (note Sec. XI.A)

Multiply $D^\dagger$ from the right by $\Gamma_5 D^{-1} \Gamma_5$ and use the exact relation of Sec. 1.5 in the form $D^\dagger = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger$, together with $\Gamma_5 D_-^\dagger \Gamma_5 = D_-$:

$$D^\dagger \Gamma_5 D^{-1} \Gamma_5 = \Gamma_5 D_-^{-1} D\, \Gamma_5 D_-^\dagger\, \Gamma_5\, D^{-1} \Gamma_5 \tag{8.1}$$

$$= \Gamma_5 D_-^{-1} D\, D_-\, D^{-1}\, \Gamma_5 \tag{8.2}$$

$$= \Gamma_5 D_-^{-1} (D_- D + [D, D_-]) D^{-1} \Gamma_5 \tag{8.3}$$

$$= \Gamma_5 (1 + D_-^{-1} [D, D_-] D^{-1}) \Gamma_5. \tag{8.4}$$

The commutator, with $D = D_+ - D_- Y$ and $[D_+, D_-] = 0$:

$$[D, D_-] = [D_+, D_-] - [D_- Y, D_-] \tag{8.5}$$

$$= -(D_- Y D_- - D_- D_- Y) = -D_- [Y, D_-] \tag{8.6}$$

$$= -D_- [Y, cW - 1] = -c\, D_- [Y, W]. \tag{8.7}$$

The factor D_- cancels against D_-^{-1}:

$$\boxed{D^\dagger \Gamma_5 D^{-1} \Gamma_5 = \Gamma_5 \,(1-N)\,\Gamma_5, \qquad N = c\,[Y,W]\,D^{-1}} \tag{8.8}$$

$[Y,W] = -\frac{1}{2}[Y,K]$ is bounded (one hop in four dimensions, one in the fifth), so N is large only on the low modes of D. There the commutator flips chirality and moves the mode to the other wall, out of the low-mode span, so N^2 is small and $(1-N)^{-1} \approx 1+N$: GCR needs only a few more steps than for the identity. Measured: 2–3 iterations more than the solve with D.

8.2 On the even sites, step by step (note Secs. XI.B–C)

The solves of HMC are on the even sites. Since Γ_5 acts within a site it commutes with π and ι, and with Eq. (2.9)

$$S^{-\dagger} = \pi D^{-\dagger} \iota \approx \pi \Gamma_5 D^{-1} \Gamma_5 \iota = \Gamma_5\, \pi D^{-1} \iota\, \Gamma_5 = \Gamma_5 S^{-1} \Gamma_5. \tag{8.9}$$

The solve of $D^\dagger w = r$:

$$r'_e = r_e - D^\dagger_{oe} (C^\dagger_o)^{-1} r_o \qquad \text{so that } S^\dagger w_e = r'_e, \tag{8.10}$$

$$w_e = (C^\dagger_e)^{-1} z \qquad \Longrightarrow\ S^\dagger (C^\dagger_e)^{-1} z = r'_e \iff K^\dagger z = r'_e, \tag{8.11}$$

$$z = \text{GCR on } K^\dagger \text{ with preconditioner } C^\dagger_e \Gamma_5 S^{-1} \Gamma_5, \tag{8.12}$$

$$S^{-1} t \approx C_e^{-1}\, \mathcal{M}\, (S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger t \qquad \text{(one cycle of } \tilde{A}_R\text{)}, \tag{8.13}$$

$$w_e = (C^\dagger_e)^{-1} z, \qquad w_o = (C^\dagger_o)^{-1} (r_o - D^\dagger_{eo} w_e). \tag{8.14}$$

The line $K^\dagger z = r'_e$ uses $K = C_e^{-1} S$: $K^\dagger = S^\dagger (C^\dagger_e)^{-1}$. The approximation for S^{-1} comes from Eq. (3.5): $S^{-1} = C_e^{-1} \tilde{A}_R^{-1} (S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger$. In the preconditioner, $(S_{\rm PV} C^{-1}_{{\rm PV},e})^\dagger t$ is the right-hand side of $\tilde{A}_R$ for the source $(t,0)$ (no odd part), so it is applied directly as one adjoint Schur factor of $D_{\rm PV}$. Each GCR step costs one cycle, as for D.

8.3 Why D_- cannot be inverted (note Sec. XI.D)

The exact relation of Sec. 1.5 contains D_-, and one could try to solve with $D^\dagger$ through D exactly by inverting D_-. Rewrite it:

$$D_- = cW - 1 = c\Big(W - \frac{1}{c}\Big), \tag{8.15}$$

c times a Wilson operator of mass $-M_5 - 1/c$. By Eq. (1.10) with $M_5 \to M_5 + 1/c$, its free eigenvalues are

$$c\Big[-M_5 - \frac{1}{c} + \sum_\mu (1-\cos p_\mu) \pm i \Big(\sum_\mu \sin^2 p_\mu \Big)^{1/2} \Big]. \tag{8.16}$$

For $M_5 = 1$, $c = 0.5$ the real part of the bracket runs from -3 to 5; near real part 0 the imaginary parts have both signs. The spectrum surrounds the origin. A Krylov method needs a polynomial p with $p(0) = 1$ that is small on the spectrum, and on a set around the origin no low-degree polynomial does that (by the maximum principle $|p|$ on a curve around 0 is at least $|p(0)| = 1$). Measured: GCR(8) on D_- stops at $|r|^2/|b|^2 \approx 0.74$ after 20000 iterations. The preconditioner of Sec. 8.1 never inverts D_-.

9 HMC

The note does not derive HMC; it reports only the measurements (note Sec. XII.G). This section goes through HMC from the path integral to the solves of the force, and then shows where the multigrid enters. Every step is written out.

9.1 What we want to compute

Lattice QCD computes expectation values in the path integral,

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}U \, \mathcal{D}\bar{\psi} \, \mathcal{D}\psi \; \mathcal{O} \, e^{-S_g[U] - \bar{\psi} D[U] \psi}, \qquad Z = \int \mathcal{D}U \, \mathcal{D}\bar{\psi} \, \mathcal{D}\psi \; e^{-S_g[U] - \bar{\psi} D[U] \psi}, \tag{9.1}$$

with U the gauge links, S_g the gauge action (here the plaquette action at $\beta = 6.0$) and ψ the quark field. The quark fields are Grassmann numbers, and their integral can be done exactly:

$$\int \mathcal{D}\bar{\psi} \, \mathcal{D}\psi \; e^{-\bar{\psi} D \psi} = \det D. \tag{9.2}$$

For two degenerate flavours (up and down) each flavour gives one $\det D$:

$$Z = \int \mathcal{D}U \; e^{-S_g[U]} \; \det D[U] \; \det D[U] \tag{9.3}$$

$$= \int \mathcal{D}U \; e^{-S_g[U]} \; \det\bigl(D^\dagger D\bigr)[U], \tag{9.4}$$

where the second line uses that $\det D$ is real, so $(\det D)^2 = \det D^* \det D = \det(D^\dagger D)$. That $\det D$ is real follows from the exact relation $D_- \Gamma_5 D^\dagger = D \Gamma_5 D_-^\dagger$ of Sec. 8, taking determinants ($\det \Gamma_5 = \pm 1$ cancels on both sides):

$$\det D_- \; \det D^\dagger = \det D \; \det D_-^\dagger \tag{9.5}$$

$$\det D_- \; (\det D)^* = \det D \, (\det D_-)^* \tag{9.6}$$

$$(\det D)^* = \det D \qquad \bigl(\det D_- \text{ is real: } D_- = cW - 1 \text{ and } \gamma_5 W \gamma_5 = W^\dagger\bigr). \tag{9.7}$$

So the problem is to generate gauge fields U with the probability

$$P[U] \;\propto\; e^{-S_g[U]} \det\bigl(D^\dagger D\bigr)[U]. \tag{9.8}$$

$\det(D^\dagger D)$ is the hard part: it couples all sites at once, and computing it directly costs $O(V^3)$.

The Pauli–Villars ratio. The domain-wall operator has L_s slices. Only the modes bound to the walls are the physical quark; the modes in the bulk of the fifth dimension are heavy, unphysical, and their determinant diverges as $L_s \to \infty$. They are removed by dividing by the determinant of the same operator at unit mass, the Pauli–Villars operator, whose wall modes are heavy too:

$$\det\bigl(D^\dagger D\bigr) \;\longrightarrow\; \frac{\det\bigl(D^\dagger D\bigr)}{\det\bigl(D_{\mathrm{PV}}^\dagger D_{\mathrm{PV}}\bigr)}. \tag{9.9}$$

This is the "Ratio" action.

9.2 Pseudofermions: a determinant as a Gaussian integral

One complex variable. For a number $a > 0$ and a complex variable $\phi = x + iy$,

$$\int \frac{dx\,dy}{\pi} \; e^{-\phi^* a^{-1} \phi} = \int \frac{dx\,dy}{\pi} \; e^{-(x^2+y^2)/a} \tag{9.10}$$

$$= \frac{1}{\pi} \Bigl(\int dx \, e^{-x^2/a} \Bigr) \Bigl(\int dy \, e^{-y^2/a} \Bigr) \tag{9.11}$$

$$= \frac{1}{\pi} \sqrt{\pi a} \, \sqrt{\pi a} = a. \tag{9.12}$$

Many variables. For a Hermitian positive matrix $A = V\Lambda V^\dagger$ (eigenvalues $a_k > 0$), change variables $\phi = V\chi$; the Jacobian of a unitary change is 1:

$$\int \mathcal{D}\phi\, e^{-\phi^\dagger A^{-1}\phi} = \int \mathcal{D}\chi\, e^{-\chi^\dagger V^\dagger V\Lambda^{-1}V^\dagger V\chi} \tag{9.13}$$

$$= \int \mathcal{D}\chi\, e^{-\sum_k |\chi_k|^2/a_k} \tag{9.14}$$

$$= \prod_k a_k = \det A. \tag{9.15}$$

So a determinant is a Gaussian integral over *bosonic* fields ϕ, the pseudofermions:

$$\det A = \int \mathcal{D}\phi\, e^{-\phi^\dagger A^{-1}\phi}. \tag{9.16}$$

The ratio. Take A such that $\det A$ is our ratio. With $H = \Gamma_5 D$ (whose determinant has the modulus of $\det D$),

$$A = H_{\rm PV}\,(H^\dagger H)^{-1}\,H_{\rm PV}^\dagger, \tag{9.17}$$

$$\det A = \det H_{\rm PV}\,\det(H^\dagger H)^{-1}\,\det H_{\rm PV}^\dagger = \frac{\det(H_{\rm PV}^\dagger H_{\rm PV})}{\det(H^\dagger H)}. \tag{9.18}$$

Note this is the *inverse* of the ratio; the action uses A^{-1} in the exponent, so

$$\int \mathcal{D}\phi\, e^{-\phi^\dagger A\phi} = \det A^{-1} = \frac{\det(H^\dagger H)}{\det(H_{\rm PV}^\dagger H_{\rm PV})} = \frac{\det(D^\dagger D)}{\det(D_{\rm PV}^\dagger D_{\rm PV})}, \tag{9.19}$$

and the pseudofermion action is

$$S_F[U,\phi] = \phi^\dagger A\phi = \phi^\dagger H_{\rm PV}(H^\dagger H)^{-1}H_{\rm PV}^\dagger\phi. \tag{9.20}$$

The joint distribution of (U,ϕ) is $e^{-S_g - S_F}$; integrating out ϕ gives back $P[U]$.

9.3 Even-odd: only K depends on the links

Factor D into block triangular matrices (LDU):

$$\begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix} = \begin{pmatrix} 1 & D_{eo}C_o^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S & 0 \\ 0 & C_o \end{pmatrix} \begin{pmatrix} 1 & 0 \\ C_o^{-1}D_{oe} & 1 \end{pmatrix}. \tag{9.21}$$

Check by multiplying out the last two factors and then the first:

$$\begin{pmatrix} S & 0 \\ 0 & C_o \end{pmatrix} \begin{pmatrix} 1 & 0 \\ C_o^{-1}D_{oe} & 1 \end{pmatrix} = \begin{pmatrix} S & 0 \\ D_{oe} & C_o \end{pmatrix}, \tag{9.22}$$

$$\begin{pmatrix} 1 & D_{eo}C_o^{-1} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} S & 0 \\ D_{oe} & C_o \end{pmatrix} = \begin{pmatrix} S + D_{eo}C_o^{-1}D_{oe} & D_{eo} \\ D_{oe} & C_o \end{pmatrix} = \begin{pmatrix} C_e & D_{eo} \\ D_{oe} & C_o \end{pmatrix}. \tag{9.23}$$

The triangular factors have determinant 1, so

$$\det D = \det S\,\det C_o \tag{9.24}$$

$$= \det\,(C_e K)\,\det C_o \qquad (K = C_e^{-1}S) \tag{9.25}$$

$$= \det C_e\,\det C_o\,\det K. \tag{9.26}$$

C_e and C_o do not depend on the links (Sec. 1.6), so they are constants and drop out of $P[U]$. Only $\det K$ is needed, on the even sites alone, half the lattice. The action becomes

$$S_F = \phi_e^\dagger H_{\rm PV}(H^\dagger H)^{-1}H_{\rm PV}^\dagger\phi_e, \qquad H = \Gamma_5 K, \quad H_{\rm PV} = \Gamma_5 K_{\rm PV}, \tag{9.27}$$

and since Γ_5 is Hermitian and unitary, $H^\dagger H = K^\dagger \Gamma_5 \Gamma_5 K = K^\dagger K$, so with $v_1 = H_{\rm PV}^\dagger \phi_e = K_{\rm PV}^\dagger \Gamma_5 \phi_e$,

$$\boxed{S_F = v_1^\dagger \, (K^\dagger K)^{-1} \, v_1} \tag{9.28}$$

This is the action of the code (Bridge++'s `Action_F_alt_Ratio_eo`).

9.4 The idea of HMC

Sampling U with $P[U] \propto e^{-S[U]}$, $S = S_g + S_F$, by small random changes is hopeless: each accepted change moves one link a little. HMC moves all links at once along a long path, and still samples $P[U]$ exactly.

Fictitious momenta. For every link introduce a momentum $\pi_\mu(x)$, a traceless Hermitian 3×3 matrix, and the Hamiltonian

$$\mathcal{H}[U, \pi] = \tfrac{1}{2} \sum_{x,\mu} \operatorname{tr} \pi_\mu(x)^2 + S[U]. \tag{9.29}$$

The joint distribution $e^{-\mathcal{H}} = e^{-\frac{1}{2}\sum \operatorname{tr} \pi^2} \, e^{-S[U]}$ is a product: the momenta are Gaussian and independent of U. Sampling (U, π) from $e^{-\mathcal{H}}$ and forgetting π samples U from e^{-S}.

One trajectory.

1. Draw π from its Gaussian and ϕ_e by the heat bath (below).
2. Evolve (U, π) for a fictitious time τ (here $\tau = 1$) by Hamilton's equations,
$$\dot{U}_\mu(x) = i \, \pi_\mu(x) \, U_\mu(x), \qquad \dot{\pi}_\mu(x) = -F_\mu(x), \tag{9.30}$$
with the force F the derivative of S with respect to U (below). Exactly solved, these conserve $\mathcal{H}$.
3. The numerical integration does not conserve $\mathcal{H}$ exactly. Accept the new U with probability $\min(1, e^{-\Delta\mathcal{H}})$, $\Delta\mathcal{H} = \mathcal{H}_{\rm end} - \mathcal{H}_{\rm start}$; otherwise keep the old U (the Metropolis test).

Why the result is exact. The test makes the method exact for any step size, provided the integrator has two properties: it is *reversible* (running it with the momenta reversed brings $(U, -\pi)$ back) and it *preserves volume* in (U, π) space. Then the proposal $(U, \pi) \to (U', \pi')$ is as likely as its reverse, and the acceptance $\min(1, e^{-\Delta\mathcal{H}})$ gives detailed balance with $e^{-\mathcal{H}}$:

$$e^{-\mathcal{H}(U,\pi)} \min\left(1, e^{-[\mathcal{H}(U',\pi') - \mathcal{H}(U,\pi)]}\right) = \min\left(e^{-\mathcal{H}(U,\pi)}, e^{-\mathcal{H}(U',\pi')}\right) \tag{9.31}$$

$$= e^{-\mathcal{H}(U',\pi')} \min\left(1, e^{-[\mathcal{H}(U,\pi) - \mathcal{H}(U',\pi')]}\right), \tag{9.32}$$

which is symmetric in the two states. A small step size only raises the acceptance.

9.5 The integrator

Leapfrog. Split a step of size ϵ into half a momentum update, a full link update, and half a momentum update:

$$\pi \leftarrow \pi - \tfrac{\epsilon}{2} F(U), \tag{9.33}$$

$$U \leftarrow e^{i\epsilon\pi} U, \tag{9.34}$$

$$\pi \leftarrow \pi - \tfrac{\epsilon}{2} F(U). \tag{9.35}$$

Each of the three updates changes only U or only π by an amount that depends on the other one, so each is volume preserving (a shear), and the product of shears is too. Reversing π and running the step again undoes it, update by update. The energy error after a trajectory is $O(\epsilon^2)$.

Omelyan. With a parameter $\lambda \approx 0.193$,

$$\pi \leftarrow \pi - \lambda\epsilon F, \quad U \leftarrow e^{i\frac{\epsilon}{2}\pi} U, \quad \pi \leftarrow \pi - (1-2\lambda)\epsilon F, \quad U \leftarrow e^{i\frac{\epsilon}{2}\pi} U, \quad \pi \leftarrow \pi - \lambda\epsilon F. \tag{9.36}$$

It is still reversible and volume preserving, and λ is chosen to make the leading error term about ten times smaller than that of leapfrog, at two force evaluations per step instead of one. $e^{i\epsilon\pi}$ is evaluated by its Taylor series (order 8 in the runs).

Nested steps. The gauge force is cheap and the fermion force expensive. The runs use two levels: the fermion force with n Omelyan steps, and within each of its link updates the gauge force with 2 steps of its own ("`number_of_steps: [n, 2]`").

How many solves. At the fermion level, consecutive momentum updates of neighbouring steps merge into one ($\lambda\epsilon + \lambda\epsilon = 2\lambda\epsilon$). Count the force evaluations of n steps:

$$\underbrace{1}_{\lambda\epsilon \text{ at the start}} + \underbrace{n}_{(1-2\lambda)\epsilon} + \underbrace{n-1}_{2\lambda\epsilon \text{ between steps}} + \underbrace{1}_{\lambda\epsilon \text{ at the end}} = 2n+1. \tag{9.37}$$

Each needs one solve with $K^\dagger K$, and the energy at the start and the end one more each:

$$\text{solves per trajectory} = 2n + 3. \tag{9.38}$$

For $n = 20$ this is 43 and for $n = 40$ it is 83, the numbers the runs report.

Step size and stability. The force grows as the quark mass falls (the low modes of K come closer to zero). If ϵ times the largest force frequency exceeds a threshold, the integrator becomes unstable and $\Delta\mathcal{H}$ explodes rather than growing smoothly. On the dynamical 16^4 configuration at $m = 0.01$, $n = 10$ gave $\Delta\mathcal{H} \approx 10^3$ (every trajectory rejected), $n = 20$ gave $|\Delta\mathcal{H}| \approx 0.05$ (acceptance 0.98), and $n = 40$ gave 0.005.

9.6 The heat bath for ϕ_e

ϕ_e must be drawn from e^{-S_F} at fixed U. Draw ξ_e Gaussian, $P(\xi_e) \propto e^{-|\xi_e|^2}$, and set

$$\phi_e = H_{\mathrm{PV}}\,(H_{\mathrm{PV}}^\dagger H_{\mathrm{PV}})^{-1} H^\dagger\,\xi_e. \tag{9.39}$$

Then $S_F = |\xi_e|^2$, so ϕ_e has the right distribution:

$$H_{\mathrm{PV}}^\dagger \phi_e = H_{\mathrm{PV}}^\dagger H_{\mathrm{PV}} (H_{\mathrm{PV}}^\dagger H_{\mathrm{PV}})^{-1} H^\dagger \xi_e = H^\dagger \xi_e, \tag{9.40}$$

$$S_F = (H_{\mathrm{PV}}^\dagger \phi_e)^\dagger (H^\dagger H)^{-1} (H_{\mathrm{PV}}^\dagger \phi_e) \tag{9.41}$$

$$= (H^\dagger \xi_e)^\dagger (H^\dagger H)^{-1} (H^\dagger \xi_e) \tag{9.42}$$

$$= \xi_e^\dagger H\, H^{-1} H^{-\dagger}\, H^\dagger \xi_e \tag{9.43}$$

$$= \xi_e^\dagger \xi_e. \tag{9.44}$$

This costs one solve with the Pauli–Villars operator ($H_{\mathrm{PV}}^\dagger H_{\mathrm{PV}}$, well conditioned: CG converges fast).

9.7 The force

The force is the change of S_F when the links change. Vary U; ϕ_e is fixed. With $\chi = \Gamma_5\phi_e$, $v_1 = K_{\rm PV}^\dagger\chi$ and $v_2 = (K^\dagger K)^{-1}v_1$, and using $\delta(M^{-1}) = -M^{-1}(\delta M)M^{-1}$:

$$\begin{aligned}
\delta S_F &= \delta\big[v_1^\dagger (K^\dagger K)^{-1} v_1\big] && (9.45)\\
&= (\delta v_1)^\dagger (K^\dagger K)^{-1} v_1 + v_1^\dagger (K^\dagger K)^{-1}\delta v_1 + v_1^\dagger\,\delta\big[(K^\dagger K)^{-1}\big]\,v_1 && (9.46)\\
&= (\delta v_1)^\dagger v_2 + v_2^\dagger\,\delta v_1 - v_1^\dagger (K^\dagger K)^{-1}\,\delta(K^\dagger K)\,(K^\dagger K)^{-1} v_1 && (9.47)\\
&= 2\,\Re\,(\delta v_1)^\dagger v_2 - v_2^\dagger\,\delta(K^\dagger K)\,v_2 && (9.48)\\
&= 2\,\Re\,(\delta K_{\rm PV}^\dagger\chi)^\dagger v_2 - v_2^\dagger(\delta K^\dagger\,K + K^\dagger\,\delta K)v_2 && (9.49)\\
&= 2\,\Re\,\chi^\dagger\,\delta K_{\rm PV}\,v_2 - 2\,\Re\,(Kv_2)^\dagger\,\delta K\,v_2. && (9.50)
\end{aligned}$$

So one force evaluation needs exactly one solve, $v_2 = (K^\dagger K)^{-1}v_1$; the rest are multiplications. δK comes from the hops in S alone (C_e does not depend on the links): for a variation of one link, only the hops through that link contribute, and the two terms above become a local outer product of v_2, Kv_2 and χ at the two ends of the link. With stout smearing the operator sees the smeared links; the force is taken with respect to them and carried back to the thin links by the chain rule through the smearing steps.

9.8 How precisely to solve

Energy. $\mathcal{H}$ at the start and the end enters $\Delta\mathcal{H}$ directly, and an error there biases the Metropolis test. These solves are done to $|r|^2/|b|^2 < 10^{-24}$.

Force. An error in the force only changes the trajectory, which the test corrects, *as long as the integrator stays reversible*. A solve from a zero initial guess is a fixed function of the links, so reversibility holds to its tolerance; 10^{-16} is ample. (Starting the force solves from the previous solutions, `number_of_initial_guess`, saves iterations but breaks reversibility at the level of the tolerance; it is off by default.)

9.9 Solving $(K^\dagger K)^{-1}$

With CG. $K^\dagger K$ is Hermitian positive, so CG solves it directly; each iteration costs two applications of K. Mixed precision runs CG in single precision with corrections in double. With Neff's α, Eq. (2.20) gives $K = Q_e K^{(\alpha)} Q_e^{-1}$ and

$$\begin{aligned}
K^\dagger K &= Q_e^{-1} K^{(\alpha)\dagger} Q_e\, Q_e K^{(\alpha)} Q_e^{-1} && (9.51)\\
&= Q_e^{-1}\,(Q_e K^{(\alpha)})^\dagger (Q_e K^{(\alpha)})\,Q_e^{-1} && (9.52)\\
&= Q_e^{-1}\,(KQ_e)^\dagger (KQ_e)\,Q_e^{-1} \qquad (Q_e K^{(\alpha)} = KQ_e), && (9.53)\\
(K^\dagger K)^{-1} &= Q_e\big[(KQ_e)^\dagger (KQ_e)\big]^{-1} Q_e, && (9.54)
\end{aligned}$$

so CG runs on the normal equation of KQ_e, better conditioned, while the action and the force stay those of K (on 24^4 at $m = 0.001$: 22% fewer iterations).

With the multigrid: two solves. The multigrid solves with D and $D^\dagger$, not with $K^\dagger K$. Invert the product, with $K = C_e^{-1}S$:

$$\begin{aligned}
(K^\dagger K)^{-1} &= K^{-1}K^{-\dagger} && (9.55)\\
&= (C_e^{-1}S)^{-1}\big[(C_e^{-1}S)^\dagger\big]^{-1} && (9.56)\\
&= S^{-1}C_e\,C_e^\dagger S^{-\dagger}. && (9.57)
\end{aligned}$$

So $z = (K^\dagger K)^{-1} r$ is two solves:

$$w = C_e^\dagger S^{-\dagger} r = C_e^\dagger \pi D^{-\dagger} \iota\, r \qquad \text{(with } D^\dagger\text{, Sec. 8)}, \tag{9.58}$$

$$z = S^{-1} C_e\, w = \pi D^{-1} \iota\, C_e w \qquad \text{(with } D\text{)}. \tag{9.59}$$

Each solve stops at 10^{-10}, while the normal equation needs 10^{-16} or 10^{-24}. So the two solves are repeated on the residual of the normal equation (iterative refinement):

$$r^{(k)} = b - K^\dagger K z^{(k)}, \tag{9.60}$$

$$z^{(k+1)} = z^{(k)} + K^{-1} K^{-\dagger} r^{(k)}, \tag{9.61}$$

each sweep gaining about ten digits, at most eleven sweeps. With α the solve with D uses $D^{-1} = QD^{(\alpha)-1}$ (the left form $\tilde{A}_L^{(\alpha)}$); the solve with $D^\dagger$ is unchanged, since $D^{(\alpha)\dagger} = QD^\dagger$ only rescales its right-hand side.

When the sweeps stall. If the sweeps stop gaining (the coarse space no longer fits the links, as after large link changes), the solve is repeated from the current z with $\mathcal{P}$ and the coarse operators rebuilt for the current links, then also after a bootstrap; if it still stalls, CG finishes it from the current z, with a warning. A trajectory is never lost to the multigrid.

9.10 What the multigrid keeps along a trajectory

The links change a little at every step of the integrator, and the setup is expensive, so the multigrid is not rebuilt for every solve.

- The test vectors y are kept from solve to solve (on the host between solves) and improved by a bootstrap, from the old y, when the pseudofermions are drawn; the filter runs only for the first trajectory.
- $\mathcal{P}$, the coarse operators and the top of the smoother's interval are kept while the links move and rebuilt for the current links when the GCR iterations per solve have grown by half (`rebuild_iteration_growth` $= 1.5$) or after a bootstrap.
- The cycle is made for each solve from these parts and dropped after it.

A kept coarse space is the coarse space of slightly different links. It is still a good preconditioner; GCR on the operator of the current links keeps the solve exact.

9.11 Memory (24^4, $m = 0.001$, 20 vectors, levels below, A30)

part	GB
HMC itself (links, smeared levels, momenta, pseudofermion, operators)	3.3
multigrid kept between solves ($\mathcal{P}$ 4.75)	6.6
fields of the sweeps	2.6
the solve with $D^\dagger$ on top (GCR, Schur factors, cycle)	6.4
peak (live)	19.1

Mixed CG needs about 9 GB on the same lattice.

9.12 What the measurements say so far

- On a thermalized dynamical 16^4 configuration (60 trajectories at $m = 0.05$, then 40 at $m = 0.01$ from the quenched one) the multigrid reproduces the trajectories of CG: the plaquette to 10^{-11}, $\Delta\mathcal{H}$ to three digits. This checks that it solves the right equation.
- There CG needs about 300 iterations per solve at $m = 0.01$ and at $m = 0.001$ alike: on a lattice this small the lowest eigenvalues are held up by the finite volume, not by the mass. The multigrid needs about 55 iterations, but one of them (a whole cycle) costs about 24 CG iterations, so it is 3–4 times slower per trajectory. Tuning gains at most 6%.
- On the quenched 24^4 configuration CG needs about 6400 iterations per solve and the multigrid wins, but a trajectory of length 1 there blows up ($\Delta\mathcal{H} \approx 4 \times 10^9$), so it is no test of HMC. There are no data yet that show the multigrid faster in HMC; that needs a volume where CG takes more than about 1300 iterations per solve.

10 Map to the code

step	where
C, C^{-1}, hops, Γ_5, Q on the device	`src/backend/cuda_bridge/mdwf.rs`; `cuda_bridge/include/fermion/`
even-odd forms, $\tilde{A}_R$, $\tilde{A}_L$ (`PvRatio`, `SchurFactor`, `AfterQ`)	`src/fermion/even_odd.rs`
GCR, FGMRES, BiCGStab, MINRES, CG	`src/solver/`
Chebyshev smoother, two-grid cycle	`src/solver/preconditioner/multigrid/cycle.rs`; `src/backend/cuda_bridge/multigrid.rs`
factorized correction	`src/solver/preconditioner/multigrid/correction.rs`; `src/backend/cuda_bridge/correction.rs`
$\mathcal{P}$: fold, restrict, prolong, CholeskyQR2	`src/backend/cuda_bridge/coarse.rs`; `cuda_bridge/include/multigrid/fold.cuh`
coarse operators by probing	`CudaMdwf::coarse_blocks` in `src/backend/cuda_bridge/mdwf.rs`
levels below	`src/backend/cuda_bridge/levels.rs`
filter, bootstrap	`low_modes` in `src/solver/preconditioner/multigrid/mdwf_even_odd.rs`; `bootstrap` in `src/backend/cuda_bridge/multigrid.rs`
solve with $D^\dagger$	`adjoint_by_gamma5` in `src/fermion/even_odd.rs`
HMC: two solves, sweeps, retries	`src/hmc/dwf_multigrid.rs`; `src/backend/cuda_bridge/hmc_multigrid.rs`
HMC: action, force, CG with α, CG fallback	`src/hmc/even_odd.rs`; `src/hmc/solve.rs`